

Ex.No.9 Use Process Explorer to identify suspicious processes

Date:

Description

Process Explorer is a powerful Windows tool that provides detailed information about running processes, allowing users to monitor, troubleshoot, and detect suspicious activities. It can be used to detect potential malware or harmful processes by analyzing various aspects of running applications.

Here's a step-by-step guide on how to use Process Explorer to identify suspicious processes on your system:

Step 1: Download and Set Up Process Explorer

1. Download Process Explorer:

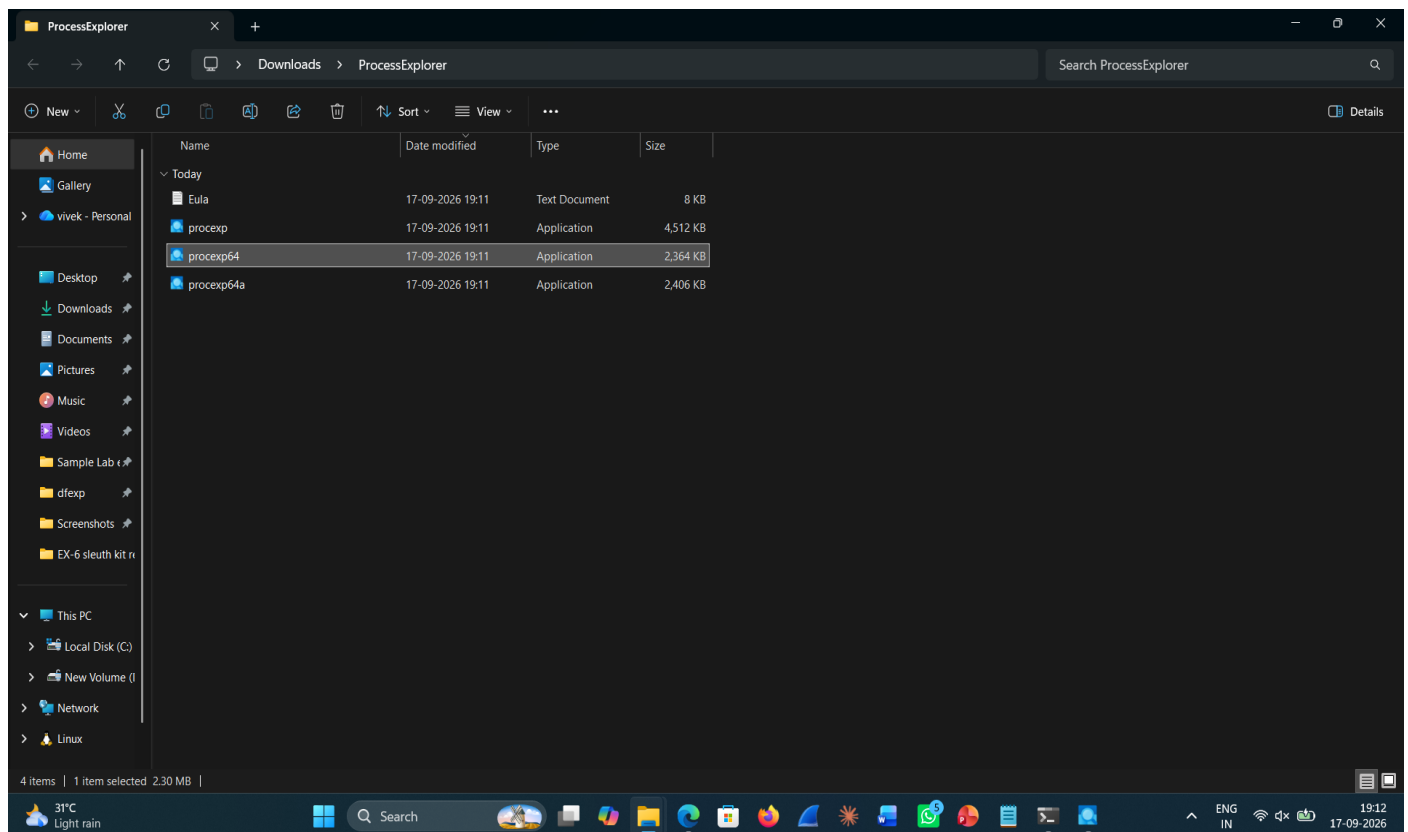
- Go to the official Microsoft Sysinternals website and download Process Explorer: [Download Process Explorer](#).

2. Extract the Program:

- Once downloaded, extract the ZIP file to a folder.

3. Run Process Explorer:

- Open the folder where you extracted the files and run `procexp64.exe` (64-bit) or `procexp.exe` (32-bit) as Administrator by right-clicking and selecting "Run as Administrator."



When Process Explorer starts, you'll see an interface that lists all running processes, showing their hierarchy, memory usage, CPU usage, and much more.

- The main window shows a tree of all running processes, with each child process listed under its parent.
- Processes are color-coded to indicate their status:
 - **Pink:** Suspended processes.
 - **Light Blue:** Processes running under the same user as you.
 - **Dark Blue:** Services or processes running under system accounts.
 - **Green:** New processes (appearing and disappearing quickly may be suspicious).
 - **Red:** Processes that have just exited.

- **Various columns show information such as Process ID (PID), CPU usage, Memory usage, and more.**

Now that you are familiar with the interface, follow these steps to investigate potentially suspicious processes:

- Scan the list of running processes and identify any names that you don't recognize. Some malware disguises itself with legitimate-looking names, but odd names can be a red flag.
- Legitimate processes should typically be from trusted sources (e.g., Microsoft, Intel, Adobe, etc.).

Process Explorer - Sysinternals

Process	CPU	Private Bytes	Working Set	PID	Description	Company Name
Secure System		172 K	63,076 K	140		
Registry		5,668 K	27,852 K	184		
System Idle Process		60 K	8 K	0		
System	< 0.01	0 K	8,016 K	4		
Interrupts	0.01	0 K	0 K	n/a	Hardware Interrupts and DPCs	
smss.exe		1,164 K	1,672 K	728		
Memory Compression		876 K	228,732 K	3124		
csrss.exe	< 0.01	2,460 K	7,016 K	972		
wininit.exe		1,576 K	9,148 K	760		
services.exe	< 0.01	7,172 K	21,188 K	1104		
svchost.exe	< 0.01	12,936 K	44,216 K	1276	Host Process for Windows S...	Microsoft Corporation
unsecapp.exe		1,936 K	10,944 K	3880		
WmPrvSE.exe		6,248 K	20,604 K	4256		
TaskManager.exe	< 0.01	11,024 K	1,98,352 K	8932	Windows Start Experience H...	Microsoft Corporation
svchost.exe	<	70,952 K	1,59,652 K	9040		Microsoft Corporation
msedgewebview2.exe		46,608 K	1,39,412 K	10852	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe		2,916 K	16,844 K	10852	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe		13,408 K	44,592 K	5996	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe		78,352 K	76,896 K	7792	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe		8,700 K	23,900 K	3704	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe		57,984 K	98,668 K	8768	Microsoft Edge WebView2	Microsoft Corporation
WidgetBoard.exe		5,744 K	69,308 K	9378		Microsoft Corporation
RuntimeBroker.exe		7,672 K	52,944 K	8656	Runtime Broker	Microsoft Corporation
WidgetService.exe		5,924 K	34,752 K	10016	WidgetService.exe	Microsoft Corporation
LockApp.exe	Susp...	70,356 K	1,60,044 K	11112	LockApp.exe	Microsoft Corporation
WidgetBoard.exe		8,660 K	44,388 K	5604	Runtime Broker	Microsoft Corporation
MicrosoftStartFeedProv...		13,776 K	54,620 K	12156		Microsoft Corporation
RuntimeBroker.exe		3,116 K	22,452 K	14888	Runtime Broker	Microsoft Corporation
FileCoAuth.exe		15,534 K	63,788 K	6736	Microsoft OneDrive File Co-A...	Microsoft Corporation
ApplicationFrameHost.exe		95,352 K	67,104 K	8664	Application Frame Host	Microsoft Corporation
SystemSettings.exe	Susp...	13,89.32 K	3,472 K	5672	Settings	Microsoft Corporation
UserOOBEBroker.exe		1,892 K	11,868 K	2094	User OOBE Broker	Microsoft Corporation
RuntimeBroker.exe		6,624 K	30,304 K	5964	Runtime Broker	Microsoft Corporation
AppxActions.exe		12,532 K	44,236 K	8552	AppxActions	Microsoft Corporation
ShellExperienceHost.exe	Susp...	43,772 K	1,14,672 K	4524	Windows Shell Experience H...	Microsoft Corporation
RuntimeBroker.exe	< 0.01	3,692 K	29,280 K	14992	Runtime Broker	Microsoft Corporation
OpenConsole.exe		2,744 K	15,152 K	11024	Console Window and PTY H...	Microsoft Corporation
Windows Terminal.exe		66,512 K	1,21,984 K	1544	Windows Terminal Host	Microsoft Corporation
WmPrvSe.exe		11,356 K	26,096 K	11192		
WinDefender.exe		2,724 K	13,020 K	10352	Windows Defender SmartScr...	Microsoft Corporation
RuntimeBroker.exe		7,848 K	10,504 K	1364		Microsoft Corporation
WUDFHost.exe		7,848 K	24,716 K	1440		
svchost.exe	< 0.01	8,332 K	18,872 K	1501	Host Process for Windows S...	Microsoft Corporation
svchost.exe		3,552 K	13,468 K	1500	Host Process for Windows S...	Microsoft Corporation
svchost.exe		3,688 K	19,888 K	1690	Host Process for Windows S...	Microsoft Corporation

CPU Usage: 3.93% Commit Charge: 47.42% Processes: 222 Physical Usage: 45.69%

2. Verify Digital Signatures:

- Right-click any suspicious process and select Properties.
- Go to the Image tab and check if the process has a valid Digital Signature.
- Legitimate applications from trusted developers will have valid signatures. Processes without signatures or with invalid signatures are suspicious.

3. Check the Path of the Process:

- In the Properties window of the suspicious process, look under the Image tab for the Path.
- Verify if the process is located in a legitimate directory. For example, legitimate system processes should reside in C:\Windows\System32.
- If the process is running from a temporary folder, user folder, or a random directory, it might be malware.

4. Monitor CPU, Memory, and Disk Usage:

- Look at the CPU, Memory, and Disk usage columns.
- A process that is using an unusual amount of resources without explanation could be malicious. For example, high CPU usage on a process like svchost.exe can indicate malware.

5. Check Process Description and Company Name:

- Look at the Description and Company Name columns in Process Explorer.
- Legitimate processes usually provide clear descriptions and are tied to well-known companies.
- Lack of information or suspicious company names should prompt further investigation.

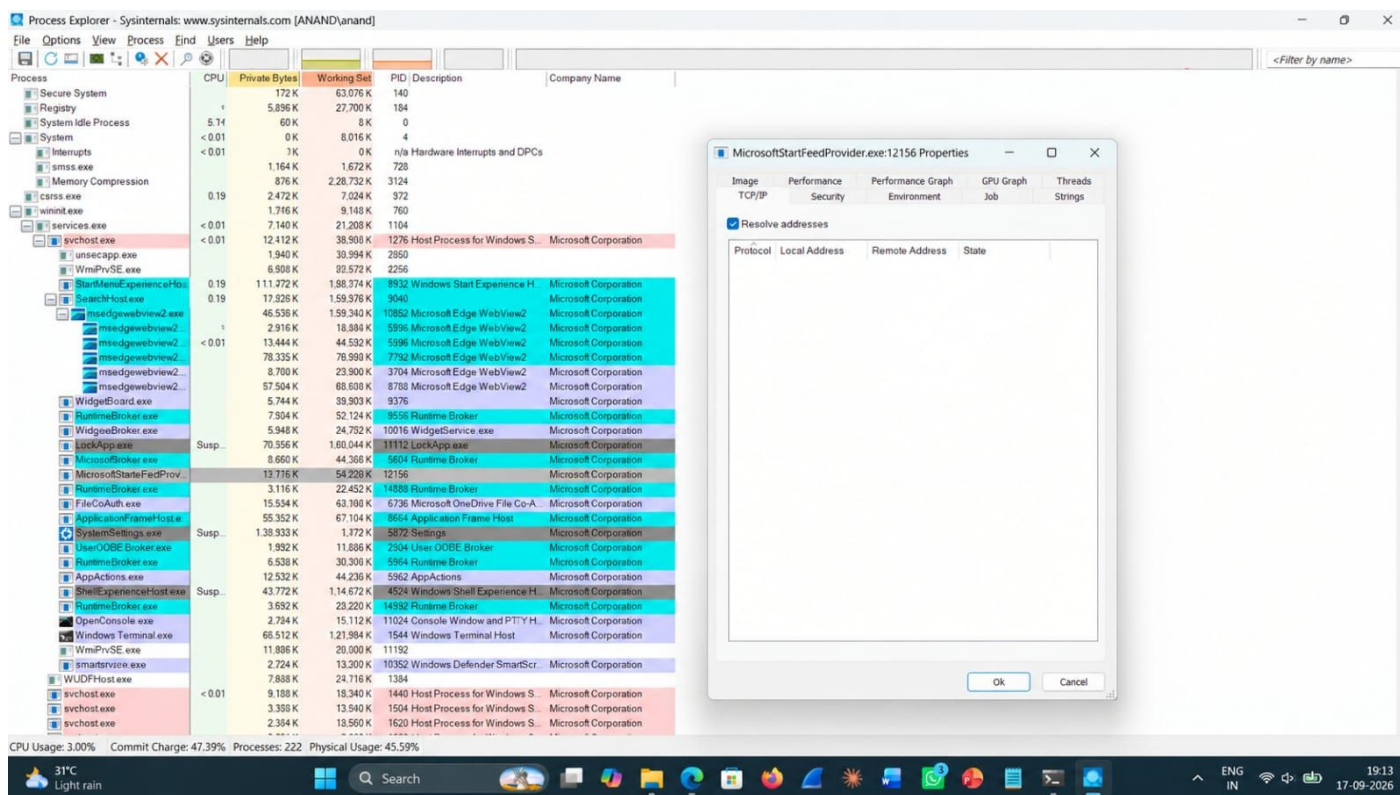
The screenshot displays the Process Explorer application window. The main pane shows a list of processes with columns for CPU, Private Bytes, Working Set, PID, Description, and Company Name. The 'svchost.exe' process is highlighted. A right-click context menu is open, and the 'Properties' option is selected. The 'MicrosoftStartFeedProvider.exe:12156 Properties' dialog box is open, showing the 'Image File' tab. The 'Image File' tab displays the following information:

- Image File: (Verified) Microsoft Corporation
- Version: 1.380.2.0
- Build Time: Wed Jul 22 08:58:44 2026
- Path: C:\Program Files\WindowsApps\Microsoft.StartExperiencesApp_1.380.2.0_x64
- Command line: "C:\Program Files\WindowsApps\Microsoft.StartExperiencesApp_1.380.2.0_x64"
- Current directory: C:\Windows\System32
- Autostart Location: n/a
- Parent: svchost.exe(1276)
- User: ANAND/anand
- Started: 17:42:17 17-09-2026
- Image: x64
- Comment:
- VirusTotal: Submit
- Data Execution Prevention (DEP) Status: Enabled (permanent)
- Address Space Load Randomization: High-Entropy, Bottom-Up
- Control Flow Guard: Enabled
- Enterprise Context: N/A
- Stack Protection: Disabled

The dialog box also includes buttons for 'Verify', 'Bring to Front', 'Kill Process', 'OK', and 'Cancel'.

6. Check the Process Network Activity:

- Some malware communicates with external servers. You can inspect a process's network connections:
 - Right-click on the process and select Properties, then go to the TCP/IP tab to see network activity associated with the process.
 - Unexpected or unexplained external connections can indicate a malicious process.



Step 4: Perform a Google Search on Suspicious Processes

If you find a process that looks suspicious or unfamiliar:

1. Search online: Perform a quick Google search using the process name (e.g., processname.exe) to find out whether the process is known to be safe or malicious.
2. Check Malware Databases: Use online databases such as VirusTotal or ProcessLibrary to verify whether the process is known for being part of malware.

Step 5: Kill or Suspend Suspicious Processes

If you find a process that you believe is malicious:

1. Kill the Process:
 - Right-click the suspicious process and select Kill Process to terminate it immediately.
2. Suspend the Process:
 - If you are unsure and want to prevent the process from causing damage while you investigate further, right-click and select Suspend.

3. Remove the Source File:

- Go to the location of the file (using the Path from the Properties window) and delete the executable file if confirmed to be malware.
- Be cautious, as some malware might prevent you from deleting it. In such cases, you may need to boot into Safe Mode or use a specialized tool.

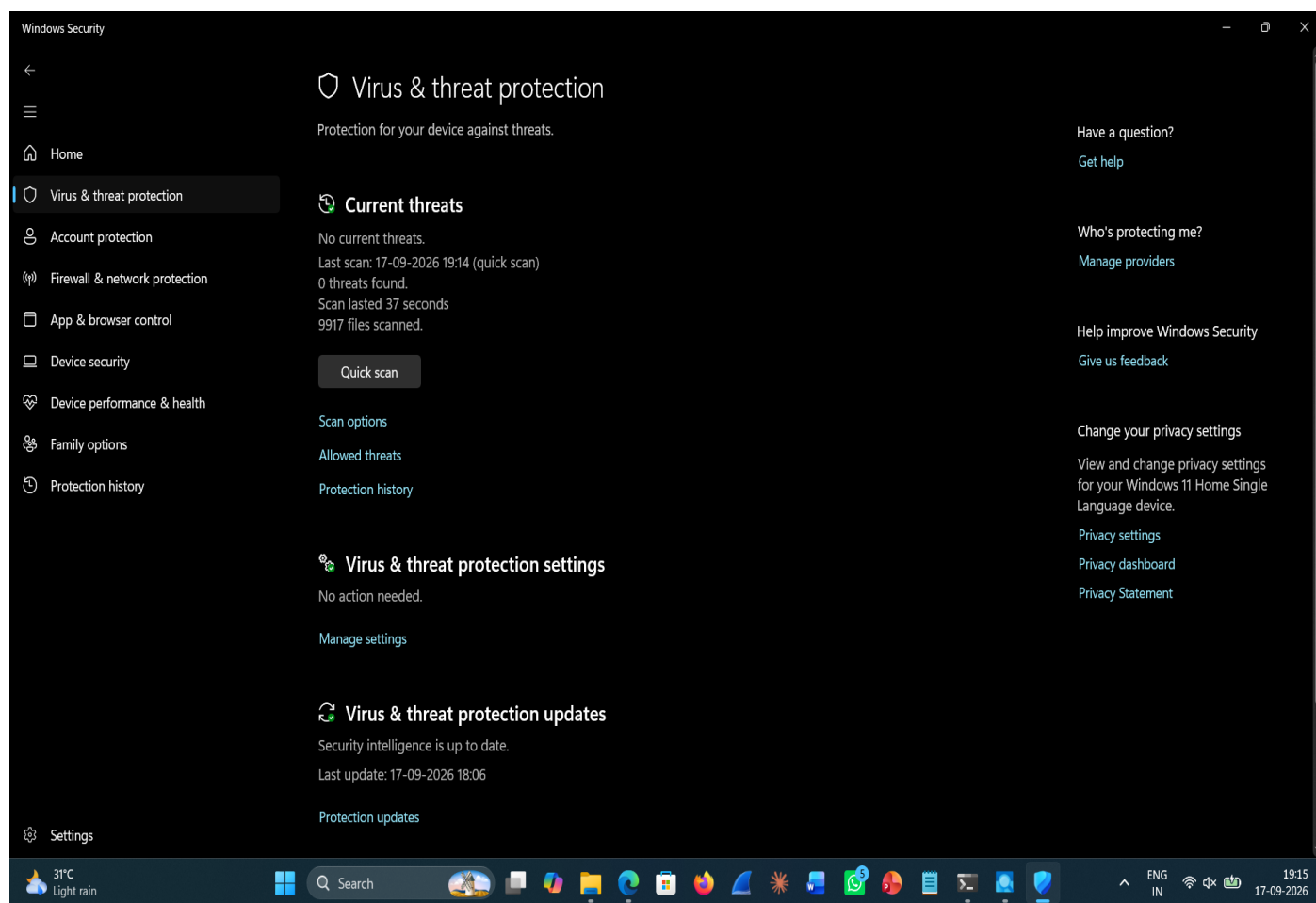
Step 6: Scan Your System

After terminating any suspicious processes, it's a good idea to:

- Run a full antivirus scan using trusted antivirus software.
- Use Malware Removal Tools like Malwarebytes or Windows Defender to perform an in-depth scan of your system.
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Example of Identifying a Suspicious Process

1. You notice a process called randomname123.exe using an unusually high amount of CPU.
2. Right-click it, select Properties, and find it is running from a suspicious folder, not the system folder.
3. You check the TCP/IP tab and see it is making external connections to unknown IP addresses.
4. A quick search shows that this process is often associated with a specific type of malware.
5. You kill the process and remove the executable file from the system.



Rubrics:

Criteria & Marks Assigned	Mark Allotted	Mark Awarded
1. GitHub Activity & Submission Regularity	3	
2. Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

Result:

Running processes were successfully monitored and analysed using Process Explorer. Process details such as digital signatures, file paths, resource usage and network activity were examined to identify potentially suspicious processes.